

# Tan Dai Ngo

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## EDUCATION

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**University of Victoria** | MEng, Applied Data Science (Co-op) *Victoria, BC | Sep 2025 - Dec 2027*

- **Courses:** Algorithms and Data Models, Systems for Massive Datasets, Information Theory. **GPA:** 8.78 / 9.00.

**University of Chicago** | MS, Computer Science *Chicago, IL, USA | Sep 2020 - Mar 2022*

- **Courses:** Distributed Systems, Databases, Applied Software Engineering, Networks.

**University of Washington** | BS, Economics *Seattle, WA, USA | Sep 2015 - Aug 2019*

- **Minors:** Informatics, Applied Mathematics. **Certificate:** Quantitative Managerial Economics. **GPA:** 3.57 / 4.00.

## EXPERIENCE

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**Software Developer** | T-Mobile (via BeaconFire Inc.) *Bellevue, WA, USA | Jun 2022 - Apr 2024*

- Built and maintained REST endpoints across 5+ Roaming Business System microservices in Java and Spring Boot.
- Kept Cassandra query response times under 3 seconds across the roaming business system.
- Automated reprocessing of rejected roaming tests with weekly Kafka retries, cutting manual effort by 80%.
- Traced 20+ cross service defects to their root cause and deployed via Jenkins across 50+ releases.

**Math and Computer Science Teacher** | APU American International Schools *Da Nang, Vietnam | Aug - Dec 2024*

- Taught Python, algorithms, object oriented design and software engineering to 50+ students.
- Explained technical material to students with no prior background, adjusting the pitch live.

## PROJECTS

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**Multi-Stage IoT Intrusion Detection** | Python, PyTorch, XGBoost *UVic Capstone, team of 7 | May - Aug 2026*

- Compared six supervised classifiers on 206,000 CIC IoT-DIAD 2024 flows under split first preprocessing.
- Selected binary XGBoost at 99.5% recall and 0.94% false positive rate against a 1% budget.
- Designed a four condition holdout that exposed session leakage, dropping PR-AUC from 0.981 to 0.125.

**Ocean Data ML Platform with RAG** | Python, Spark, Qdrant, FastAPI *Personal Project | Feb - Mar 2026*

- Ingested 10.8M NOAA and Ocean Networks Canada observations into layered Bronze and Silver Spark tables.
- Served model inference and plain English retrieval behind FastAPI with Qdrant and Ollama, at hit@k 0.90.
- Containerized the FastAPI service and vector store with Docker Compose for identical local runs.

**Anomaly Detection at Scale on Species Data** | Python, PySpark, scikit-learn *UVic, team of 3 | Mar - Apr 2026*

- Implemented Isolation Forest from the published algorithm rather than a library, then validated it.
- Verified linear runtime growth on 1,093,203 records using stratified samples across eight bins.
- Identified a geographic form of swamping across features and proposed two mitigations.

**Automated Crypto Trading Bot** | Python, Coinbase API, AWS, pandas *Personal Project | Feb - Jul 2025*

- Implemented a mean reversion engine that buys 1% dips and sells on 2% rises inside 60 minute windows.
- Added a second strategy that buys only when the 8 period EMA is above the 20 and price is above the 200.
- Backtested both on four months of polled Coinbase data, then shipped to EC2 through a CI/CD pipeline.

## TECHNICAL SKILLS

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**Languages & Programming:** Python, Java, object oriented programming, scripting

**Machine Learning & Evaluation:** PyTorch, XGBoost, scikit-learn, pandas, NumPy, Optuna, MLflow, classification

**Data, Retrieval & Messaging:** Spark, PySpark, ETL, Qdrant, RAG, embeddings, vector search, Cassandra, Kafka

**Backend, Cloud & CI/CD:** Spring Boot, FastAPI, REST, Docker, AWS (EC2, S3, Lambda, SNS), Jenkins deployments

**Engineering Practice:** pytest, API testing, model evaluation, experiment tracking, code review, Postman, Jira